DOCUMENTHUB

PROJECT STATUS

Report

Modules, features and architecture

Implementation inventory and next-stage feature plan

STATUS DATE

15 October 2026

ASSESSMENT BASIS

Repository source review

EXECUTIVE SUMMARY

DocumentHub has a substantial end-to-end implementation foundation: account access, document upload and management, automated text extraction and enrichment, stored summaries, vector-based

FRONTEND	BACKEND	AI / RAG	DATA
React + Vite	**Express APIs**	**OCR + retrieval**	**Supabase**

COMPLETION ASSESSMENT

- Core backend workflows are implemented in source, including authenticated document CRUD and background AI processing.
- The document library connects to upload, view/delete, summaries and document-scoped AI Q&A; search/filter/sort and grid/table views are present.
- The dashboard is still a presentation shell: its displayed counts are hard-coded placeholders, and AI is labeled Preview / Coming Soon.
- The code calls Supabase tables, Storage and vector RPC functions, but database migrations and RPC definitions are not included in this repository.

HOW TO READ THIS REPORT

-Implemented- means evidenced in source code. It does not certify a live deployment, configured Supabase schema, successful end-to-end run, or production readiness.

Modules and contents

A module-by-module view of the current implementation scope.

01 Frontend application

Core screens present

React 19 + Vite single-page app with React Router and Material UI. Includes landing/auth dialogs, dashboard shell, document library, upload form, grid/table views, category and name filters, sorting, and AI Summary / Ask AI dialogs. The dashboard metrics and recent-document panel remain placeholders.

02 Backend API and access control

Implemented in source

Express API groups authentication, document and AI endpoints. Registration hashes passwords with bcrypt; login issues a 24-hour JWT. Middleware protects document and AI routes, and document queries are scoped to the signed-in user.

03 Document ingestion and enrichment

Implemented in source

Uploads are accepted for PDF, PNG and JPEG (up to 50 MB in the upload middleware), stored in Supabase Storage, and processed asynchronously. Processing extracts text, metadata and entities, predicts a category, generates a summary, and updates processing fields.

04 RAG and AI services

Implemented; runtime config needed

OCR handles embedded-text PDFs and scanned PDF pages, plus preprocessed image OCR. Embeddings use Xenova/all-MiniLM-L6-v2. Text is chunked at 1,000 characters with 200-character overlap. RAG retrieves up to five chunks for document Q&A; semantic search retrieves up to ten matches. Ollama is the preferred answer provider with Gemini fallback.

05 Database and file storage

Supabase integration present

The backend uses Supabase for users/documents records, a documents Storage bucket, and document_chunks plus insert_document_chunk / match_document_chunks RPC calls for vector writes and retrieval. The repository does not contain SQL migrations or the RPC definitions, so schema deployment cannot be verified from source alone.

Technology references reflect package manifests and active source imports.

Features included

User-facing workflows and AI capabilities currently evidenced in the codebase.

Account registration and login

Create an account and sign in with email/password. Passwords are hashed with bcrypt; successful login returns a JWT used by protected API requests.

Personal document vault

Upload supported files, view them through a time-limited signed URL, list and delete owned documents, and persist document metadata in Supabase.

Document organization

Store category and optional description. The library provides name/category filtering, sort choices, and grid/table display modes.

PDF and image text extraction

Extract embedded PDF text; when a PDF is scanned, render pages and run OCR. Images are preprocessed with rotation, resize, grayscale, normalization and sharpening before Tesseract.

Metadata and category enrichment

Extract document type, people, dates, email/phone and supported identifiers using rules and entity parsing; classify into Identity, Education, Medical, Financial, Legal or Others.

Automatic document summaries

Generate a concise summary using the configured local Ollama model (default Gemma 3 4B), with a lightweight extractive fallback if local generation fails.

Document-scoped AI Q&A

Ask questions about a selected document. Retrieval combines stored summary/metadata and the most relevant chunks; the answer service uses Ollama first and Gemini as fallback and returns source references.

Semantic search endpoint

The authenticated backend embeds a query and searches the user's document chunks. A frontend service wrapper exists; a global semantic-search screen is not wired into the reviewed document-library page.

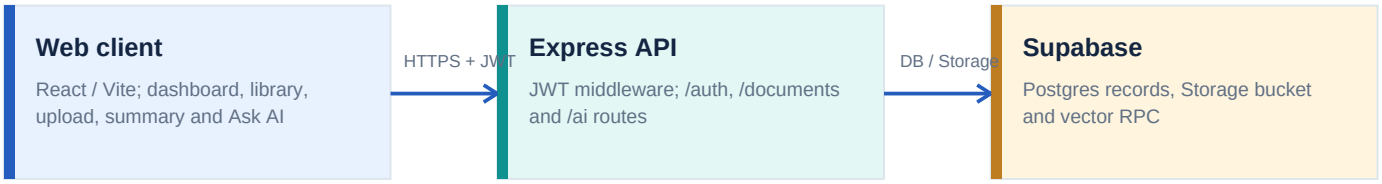
STATUS NOTE

Processing status fields and UI badges exist; confirm background workers, credentials, database functions and model availability in the deployment environment.

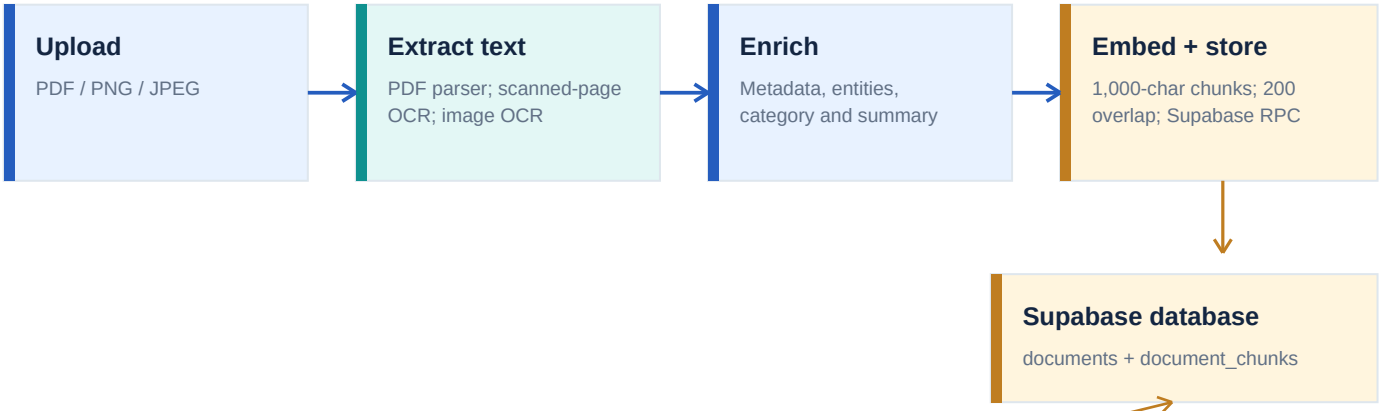
System architecture

Implemented request paths and persistence boundaries (logical view).

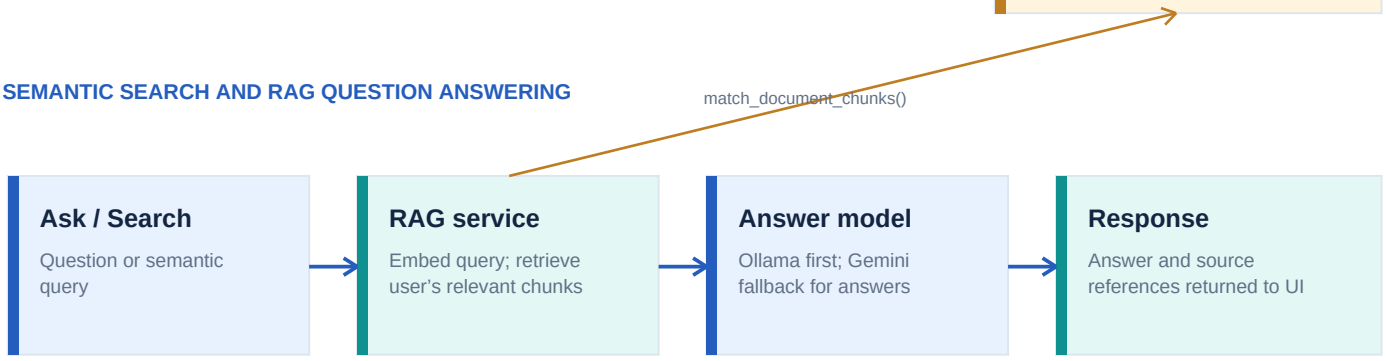
APPLICATION AND PERSISTENCE



UPLOAD PROCESSING PATH



SEMANTIC SEARCH AND RAG QUESTION ANSWERING



CONFIGURATION BOUNDARY

Supabase SQL/RPC definitions, environment secrets, Storage policies, model runtimes and deployment settings are external configuration and are not verifiable from the repository snapshot.

Next-stage feature plan

Proposed product features for the next development stage.

01

Live dashboard insights

Replace placeholder counters with live document totals, storage usage, category distribution and a recent-activity feed so users can understand their vault at a glance.

02

Whole-library semantic search

Add a dedicated search experience across all of a user's documents, showing matched passages, document names and direct links to the source files.

03

Expiry and due-date reminders

Let users review extracted dates, mark important expiry dates and receive configurable reminders for time-sensitive identity, insurance and financial documents.

04

Secure document sharing

Allow a user to create, manage and revoke controlled sharing links, with configurable expiry and read-only access for selected documents.

05

Review and export AI results

Provide a user review screen for extracted metadata and categories, plus export of a document's summary and structured fields for offline reference.

DELIVERABLE INTENT

These are proposed user-facing capabilities, not claims about current implementation. Prioritize based on user feedback and available time.